PA3 Part C — Use-Case Model (SCRUM-36)

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

Prepared: 2026-11-09. Revised 2026-11-14: delivery scopes now match the PA3 implementation (Part E is FG-01; FG-03 moves to PA4).

1. Scope and sources

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

This model covers the complete intended DebateSpeak product, including all 60 functional requirements. It is an analysis deliverable for PA3; the presence of a use case does not mean that its feature is implemented in PA3. The implemented sprint boundary remains the current roadmap and sprint scope.

Authority	Use in this model
PA3 assignment, Parts C and D	Actors, use cases, relationships where applicable; every use case proceeds to a detailed specification and UI evidence
Vision Document, §5	User-facing goals across all ten functional groups
Architecture, §§2–3 and 6–10	Actor permissions, FR-01–60, room lifecycle, consent, grounding and evaluation rules
PA3 scope and Changes	Build 1 reduction, later audio/analysis boundaries and stretch items
Roadmap	Current build sequence; older sprint-number annotations in individual FR rows do not redefine this sequence
G1 room contract, §§2–3	Two-person capacity, join idempotency, allowed state graph and API/service boundaries

The JSON catalogue is the shared index for this model, the detailed specifications and prototype screenshots. IDs remain stable when wording changes. Requirement mapping is many-to-many: several goals can implement one FR, and one goal can satisfy several related FRs.

2. Actors and authorization boundary

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

ID	Actor	Scope and permitted interaction
USER	Authenticated user	abstract: Shared authenticated identity; general actor only, not a permission to access every role.
VISITOR	Visitor	core: Registers, signs in, or requests password recovery.

ID	Actor	Scope and permitted interaction
LEARNER	Learner / Debater	core: Practises with a partner and controls own learning data.
ADMIN	Moderator / Admin	core: Performs authorized moderation and operational tasks.
EVALUATOR	Human evaluator	should: Grades only assigned, ended debates under dual sharing consent.
TEACHER	Teacher	post-mvp: Reads explicitly shared student dashboards; no class-management workflows are designed.
AUDIENCE	Audience member	post-mvp: Receives only authorized room media; never publishes or grades.
MEDIA	LiveKit Cloud	external: Routes independent participant audio tracks.
STT	STT / diarization provider	external: Transcribes authorized audio through STTAdapter.
LLM	LLM provider	external: Returns schema-validated analysis through LLMAdapter; not the authorizing actor.
SEARCH	Search / evidence provider	external: Returns evidence snippets through SearchAdapter.
PRONUNCIATION	Pronunciation provider	stretch: Optional separate assessment provider.

An actor is a role outside the DebateSpeak application boundary. Internal Language Coach, Argument Coach, Judge, Consensus Engine, ARQ worker and Grounding Guard modules are not external actors. LLM, STT, media and search providers are supporting external systems; they cannot authorize a learner action or grant consent. One person can hold multiple roles, but each operation is checked against the active role and resource authorization.

Generalization denotes shared authenticated identity only: Learner, Admin, Evaluator and Teacher specialize Authenticated user. It never grants an Evaluator administrator powers or access to an unassigned transcript. The AI opponent in a future solo mode is application behavior, never a second consenting human. The required TTS adapter for that mode has not been specified for production and must be designed before implementation.

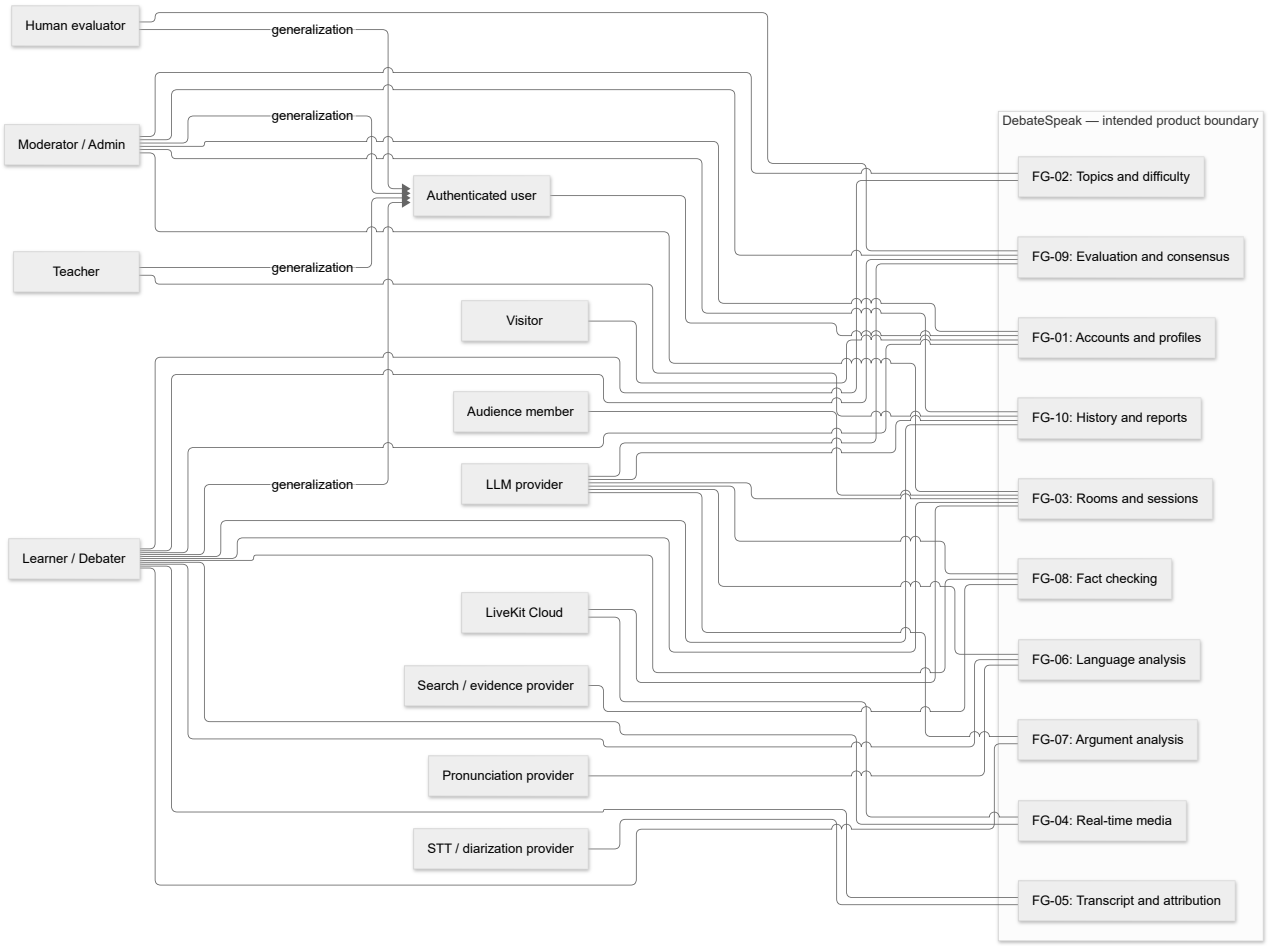
3. Notation, relations and overview

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

Mermaid flowcharts approximate UML use-case notation: rounded stadium nodes are use cases, rectangular nodes outside the boundary are actors, and a `DebateSpeak` subgraph is the system boundary. A solid undirected line is actor participation. Dotted arrows labelled `«include»` go from a base use case to mandatory reused behavior. Dotted arrows labelled `«extend»` go from the optional use case to its base, under the stated extension condition. Solid arrows labelled `generalization` point from specialized actor to general actor. Mermaid does not provide the native UML hollow-triangle actor symbol; the explicit label defines the semantics.

A signed-in session is normally a precondition, not an `include Sign in` relationship: already-authenticated users do not sign in again on every action. Chronological order alone does not establish include or extend. Supporting subcases UC-03-13 and UC-06-05 run inside other use cases; their prototype views explain visible gate results and rejected feedback, rather than add public endpoints or standalone navigation.

The overview below uses rectangular FG nodes as a coverage map, not as additional use cases. Detailed diagrams follow for each FG. Actor associations indicate participation, not unrestricted read access.



4. Relationship catalogue

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

Base or extension	Related use case	Relation and condition
UC-03-05	UC-03-13	include: Every start request executes authoritative gate validation.
UC-06-01	UC-06-05	include: Every retained language feedback item must pass citation validation.
UC-07-01	UC-06-05	include: Every retained argument critique must pass citation validation.
UC-07-02	UC-06-05	include: Every retained potential-fallacy critique must pass citation validation.
UC-02-03	UC-02-01	extend: At topic discovery, when the learner requests personalized recommendations.
UC-04-04	UC-04-02	extend: At media setup, when optional video is enabled and separately authorized.
UC-08-02	UC-04-02	extend: During LIVE audio, when the learner opts into supported live evidence overlays.

Base or extension	Related use case	Relation and condition
UC-09-06	UC-09-01	extend: At the evaluation view, when a human grade is submitted and both debaters still authorize sharing.
UC-10-05	UC-10-01	extend: At report review, when the learner requests a supported PDF export.

For UC-03-05 the start request always runs UC-03-13 against current authoritative state, even if the UI previously showed ready. UC-06-05 validates speaker and segment identity in the authorized transcript collection as well as exact quote text. It does not prove that the criticism itself is correct. Optional live video and live fact checking remain unavailable unless their feature, consent and operational contracts have been implemented. Human-grade visibility is reevaluated on each access; an earlier consent record is not an unrestricted future grant.

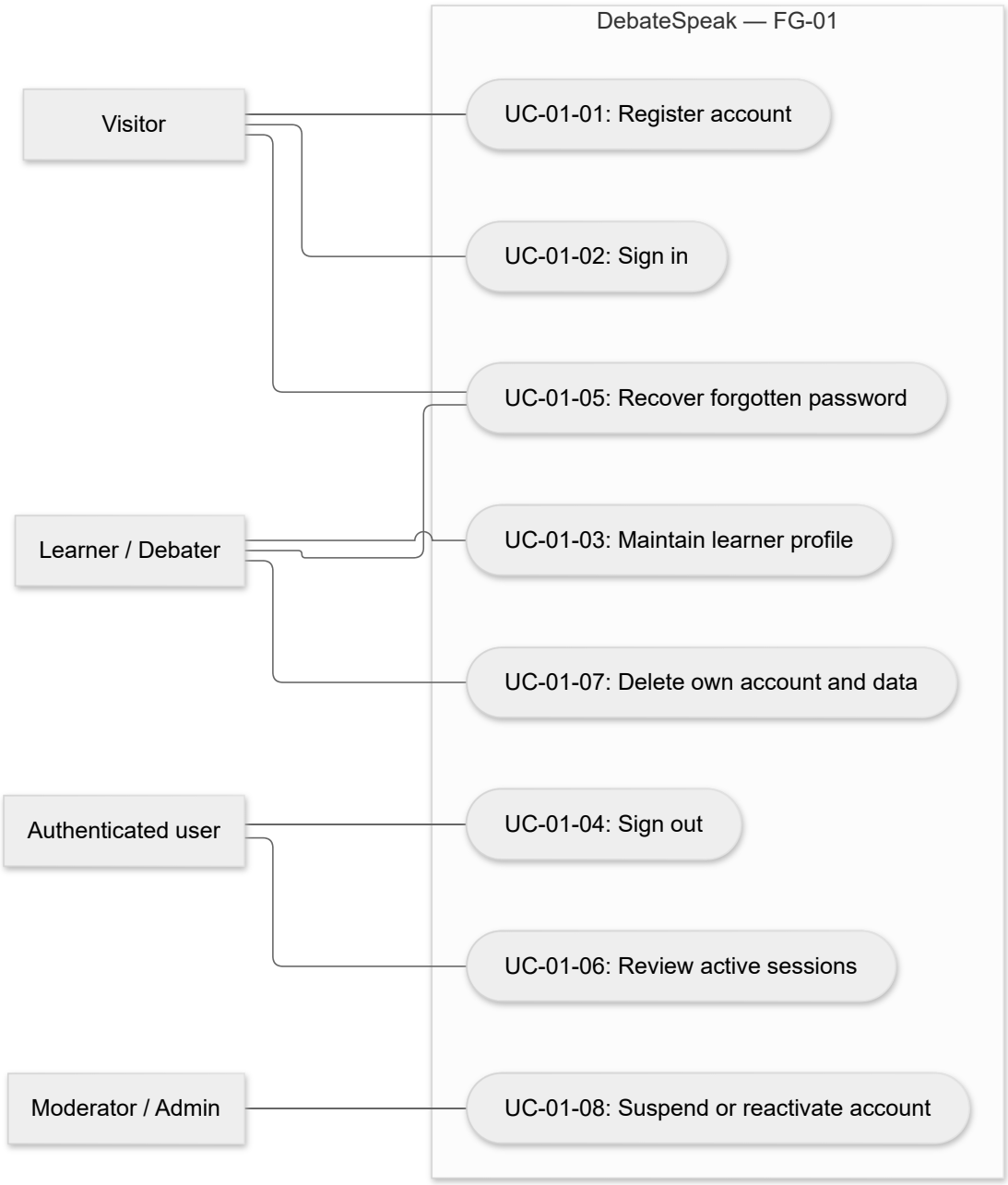
5. Functional-group diagrams and use-case inventory

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

FG-01 — Accounts and profiles

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14.

FR-01 maps to registration and sign-in. UC-01-02 also specifies the full-product refresh-session lifecycle (renewal, expiry, revocation, replay and lost responses) in specification Section 2.5. Refresh is supporting session behavior, not an additional user goal or an include from registration. PA3 implements all eight FG-01 use cases end to end as the Part E slice: access JWT in memory, rotating HttpOnly refresh cookie, profile, sessions, deletion and admin suspension, stored in PostgreSQL. The API also implements UC-01-05 recovery and email verification; the frontend keeps recovery links hidden until real email delivery is verified. Both auth flows inherit the detailed error contract in specification Section 2.4.



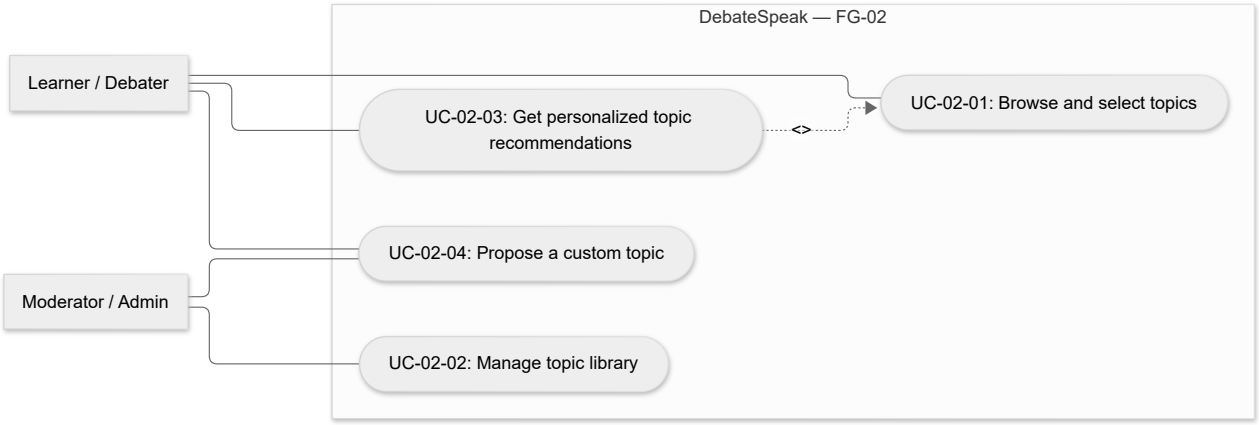
ID	User goal / supporting behavior	FRs	Delivery scope
UC-01-01	Register account	FR-01	PA3
UC-01-02	Sign in	FR-01	PA3
UC-01-03	Maintain learner profile	FR-02	PA3
UC-01-04	Sign out	FR-03	PA3
UC-01-05	Recover forgotten password	FR-03	PA3 (UI gated until email delivery is verified)
UC-01-06	Review active sessions	FR-03	PA3
UC-01-07	Delete own account and data	FR-04	PA3
UC-01-08	Suspend or reactivate account	FR-05	PA3

Account deletion immediately disables access, followed by hard deletion within seven days. A reset request must not reveal whether an email is registered. Profile CEFR is a self-assessment. In PA3 the whole of FG-01 is the

graded slice. The scheduled hard-delete job after seven days is specified but not built yet. A signed-in learner reaches UC-01-05 from Profile (Change password). The build has one administrative role, `ADMIN`.

FG-02 — Topics and difficulty

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

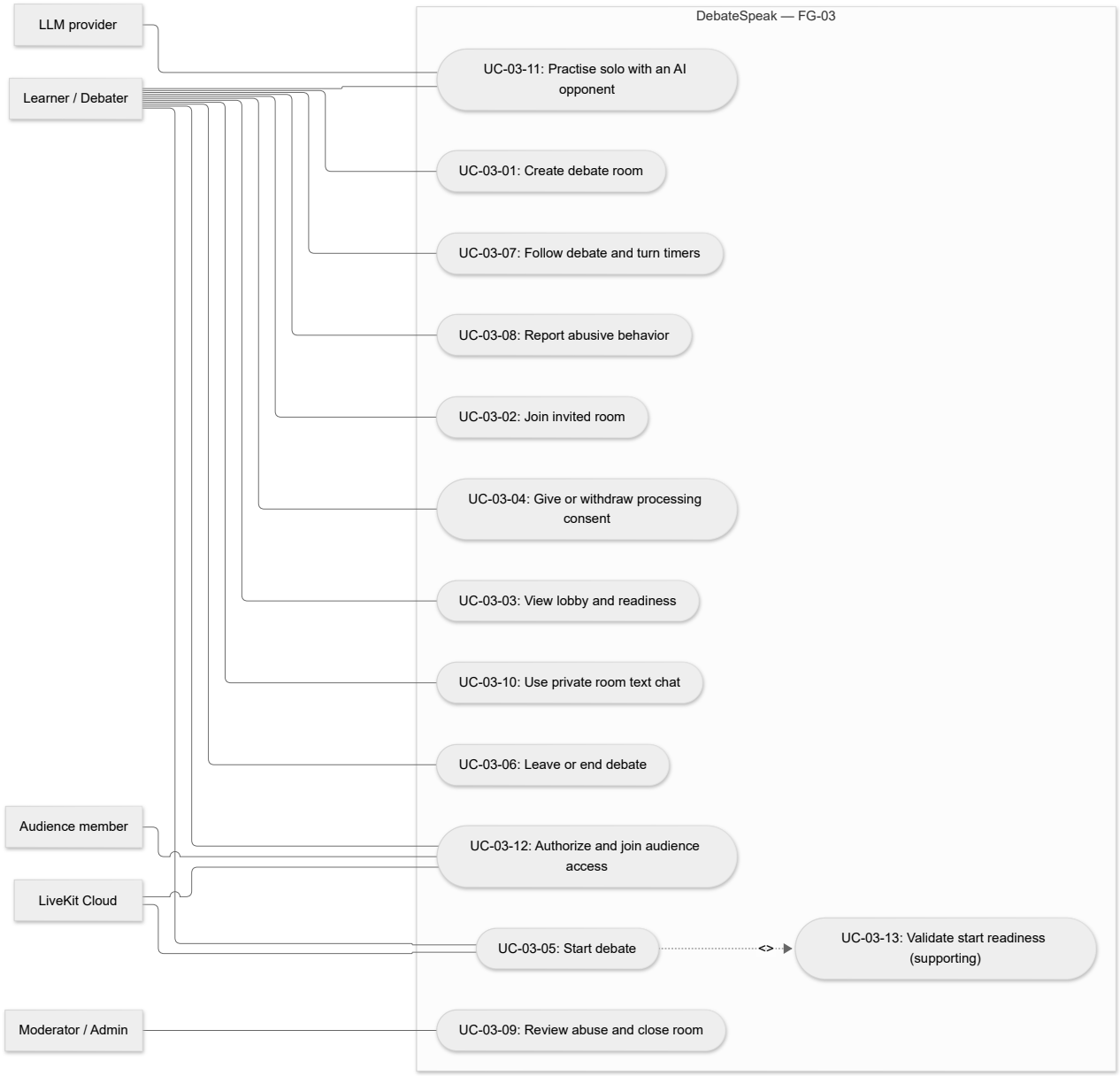


ID	User goal / supporting behavior	FRs	Delivery scope
UC-02-01	Browse and select topics	FR-06	PA3
UC-02-02	Manage topic library	FR-07	future-mvp
UC-02-03	Get personalized topic recommendations	FR-08	should
UC-02-04	Propose a custom topic	FR-09	stretch

PA3 may use the agreed seeded topic catalogue until a topic API is available. Topic recommendation is a conditional extension of discovery, not a mandatory search step. Retiring a topic must not rewrite an existing debate's history.

FG-03 — Rooms and sessions

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14.



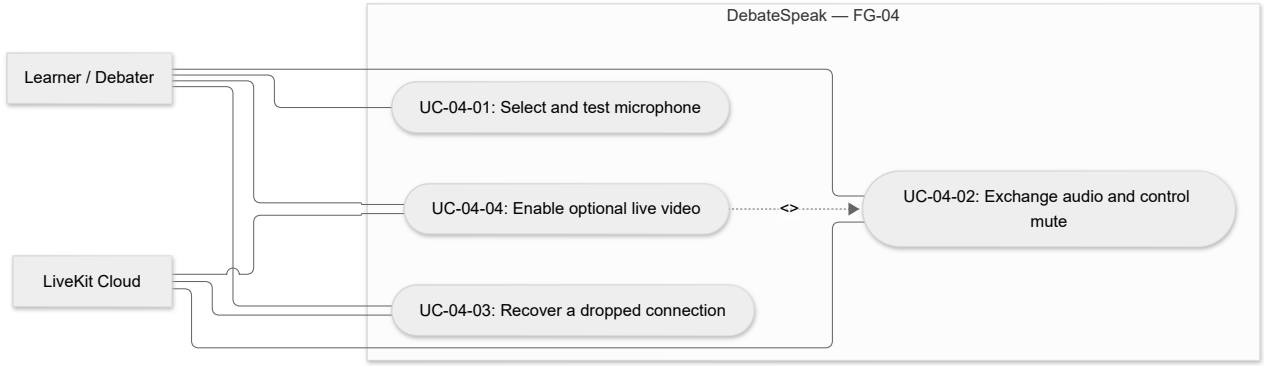
ID	User goal / supporting behavior	FRs	Delivery scope
UC-03-01	Create debate room	FR-10, FR-12	PA4 (PA3 UI preview only)
UC-03-02	Join invited room	FR-11, FR-12	PA4 (PA3 UI preview only)
UC-03-03	View lobby and readiness	FR-12	PA4 (PA3 UI preview only)
UC-03-04	Give or withdraw processing consent	FR-13	future-mvp
UC-03-05	Start debate	FR-12, FR-13, FR-19	future-mvp
UC-03-06	Leave or end debate	FR-12, FR-15	future-mvp
UC-03-07	Follow debate and turn timers	FR-14	future-mvp
UC-03-08	Report abusive behavior	FR-16	should
UC-03-09	Review abuse and close room	FR-17	should
UC-03-10	Use private room text chat	FR-58	post-mvp

ID	User goal / supporting behavior	FRs	Delivery scope
UC-03-11	Practise solo with an AI opponent	FR-59	post-mvp
UC-03-12	Authorize and join audience access	FR-60	post-mvp
UC-03-13	Validate start readiness	FR-12, FR-13, FR-19	future-mvp; supporting

Exactly two debaters count toward capacity. LOBBY becoming READY indicates membership only. Dual consent and explicit local mic tests are separate start gates. Join retries return the same membership; a third distinct user is rejected even under concurrent joins. CANCELLED and ENDED are terminal. Build 1 stops at READY; the model must not activate media or a generic status-patching endpoint.

FG-04 — Real-time media

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

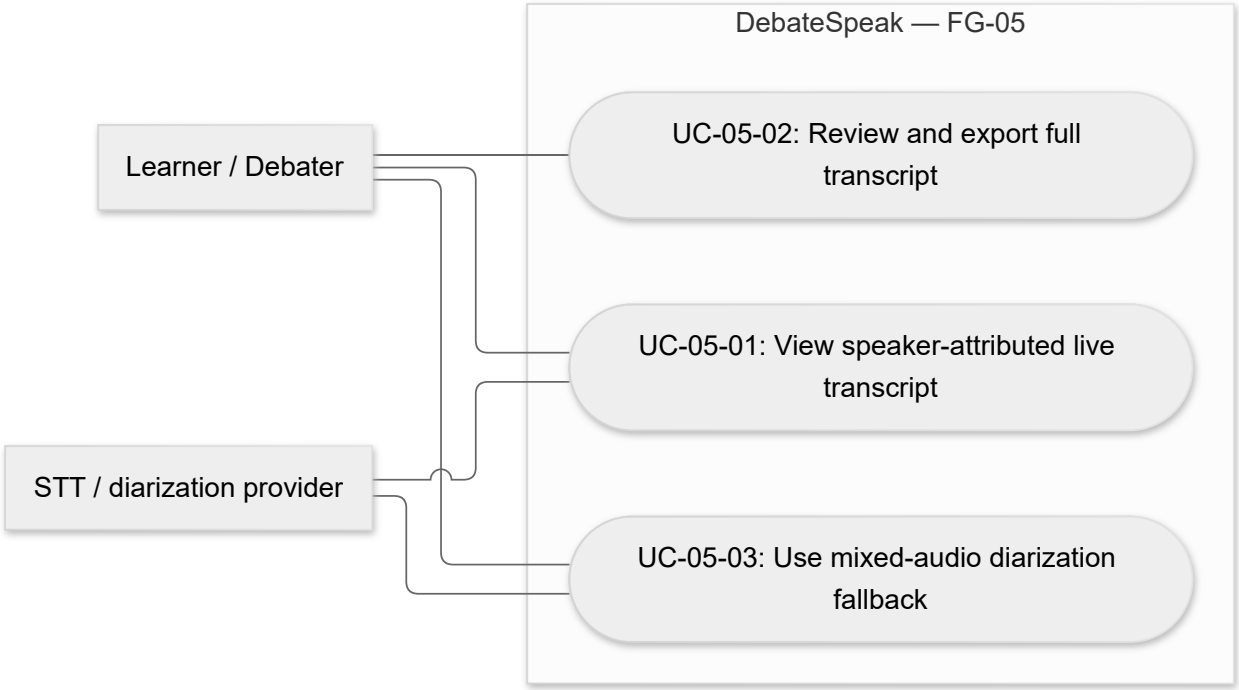


ID	User goal / supporting behavior	FRs	Delivery scope
UC-04-01	Select and test microphone	FR-19	future-mvp
UC-04-02	Exchange audio and control mute	FR-18, FR-20	future-mvp
UC-04-03	Recover a dropped connection	FR-15, FR-21	future-mvp
UC-04-04	Enable optional live video	FR-22	stretch

No network audio publication or STT starts without dual consent. The local mic test begins only by explicit action and stays local. Audio has priority over optional video, overlays and analysis. A dropped connection retains the existing identity during a visible 60-second grace period.

FG-05 — Transcript and attribution

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

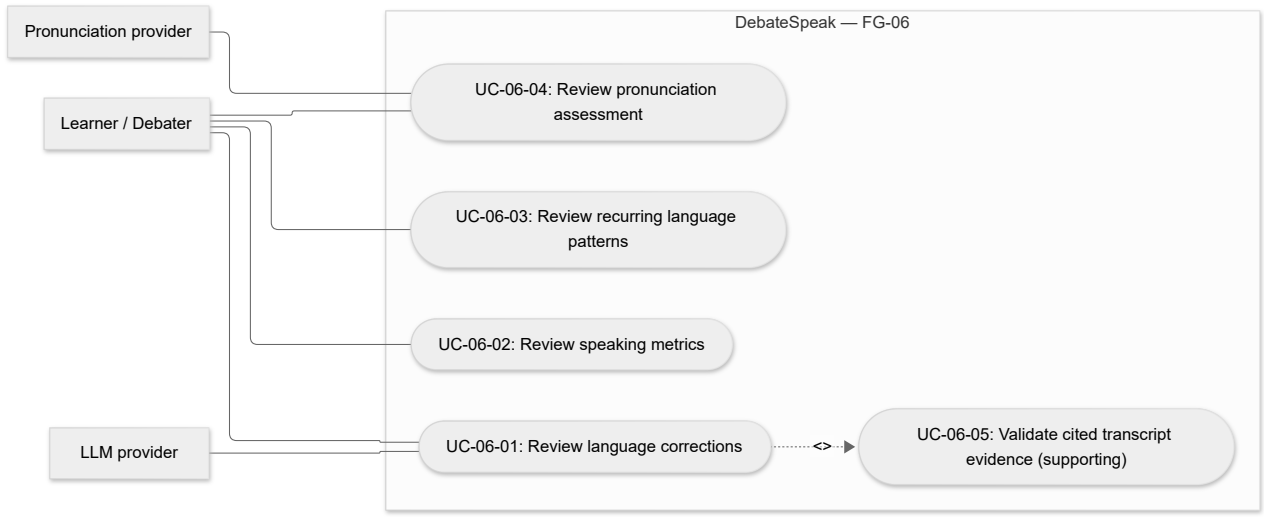


ID	User goal / supporting behavior	FRs	Delivery scope
UC-05-01	View speaker-attributed live transcript	FR-23, FR-24, FR-25, FR-26	future-mvp
UC-05-02	Review and export full transcript	FR-27	should
UC-05-03	Use mixed-audio diarization fallback	FR-28	stretch

Independent LiveKit track identity is the primary attribution source. Partial segments are replaced by final text and final segment persistence is idempotent. Low-confidence or missing text is displayed honestly. Mixed-audio diarization is an optional separately consented fallback, not permission to infer speaker identity in the normal path.

FG-06 — Language analysis

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

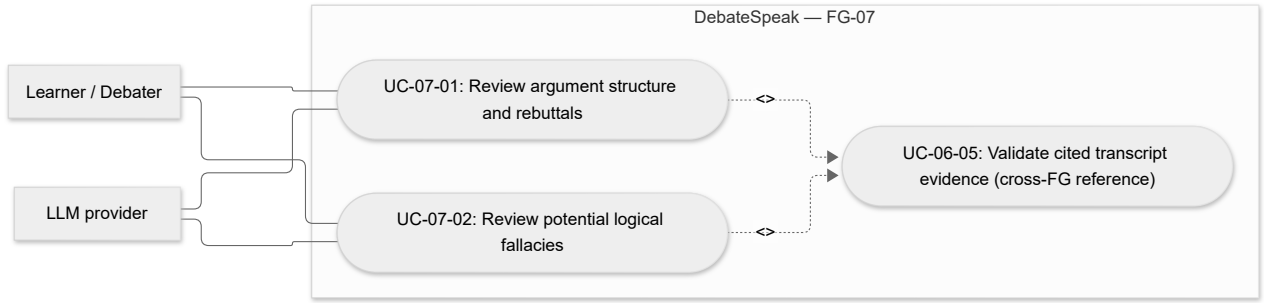


ID	User goal / supporting behavior	FRs	Delivery scope
UC-06-01	Review language corrections	FR-30, FR-31	future-mvp
UC-06-02	Review speaking metrics	FR-29	future-mvp
UC-06-03	Review recurring language patterns	FR-32	should
UC-06-04	Review pronunciation assessment	FR-33	stretch
UC-06-05	Validate cited transcript evidence	FR-30, FR-37	future-mvp; supporting

Language criticism is direct, evidence-based and paired with a correction or practice action. The grounding subcase checks exact speaker/segment quotes; uncertain STT cannot become a confirmed learner error. Deterministic metrics expose their units and missing-data limitations. Pronunciation is a separate optional provider flow.

FG-07 — Argument analysis

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).



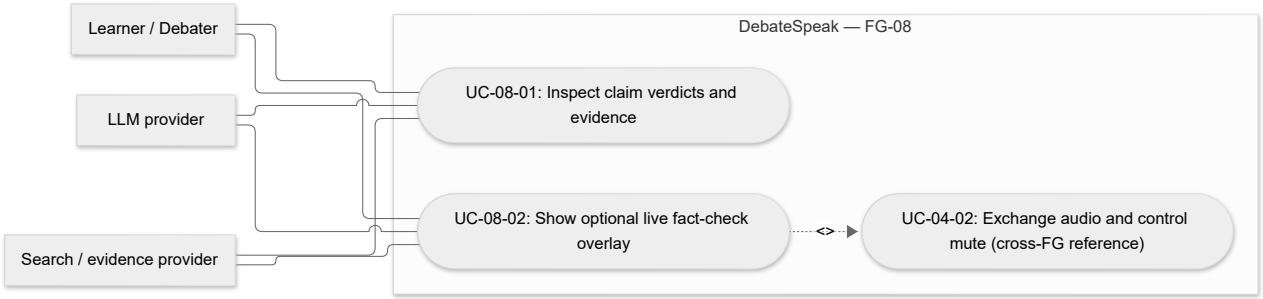
ID	User goal / supporting behavior	FRs	Delivery scope
UC-07-01	Review argument structure and rebuttals	FR-34, FR-35, FR-36	future-mvp

ID	User goal / supporting behavior	FRs	Delivery scope
UC-07-02	Review potential logical fallacies	FR-37	should

Argument quality is judged independently of accent and grammar. Potential fallacies require a quoted example and context, not a label generated without evidence. Each retained critique invokes the same deterministic citation guard as language feedback.

FG-08 — Fact checking

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

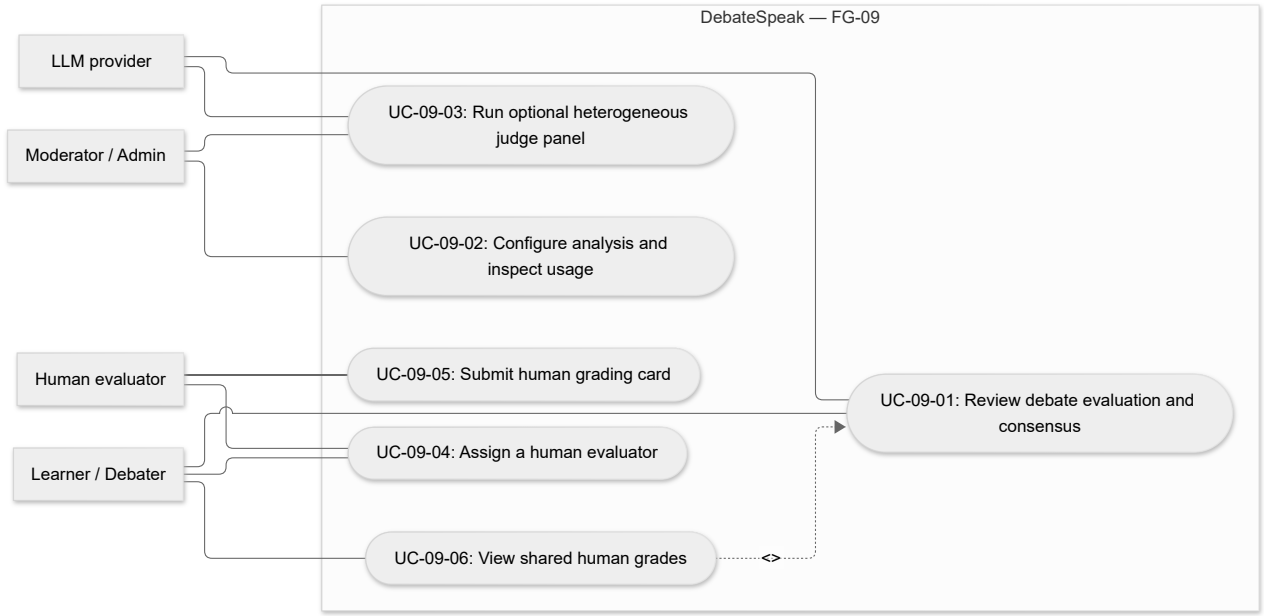


ID	User goal / supporting behavior	FRs	Delivery scope
UC-08-01	Inspect claim verdicts and evidence	FR-38, FR-39, FR-40, FR-41, FR-43	future-mvp
UC-08-02	Show optional live fact-check overlay	FR-42	stretch

Default fact-check workload is capped at eight selected claims. External evidence is untrusted data. Five verdicts and uncertainty are preserved; disagreement between Evidence Evaluator and Skeptic remains visible. A live overlay is optional and cannot block audio.

FG-09 — Evaluation and consensus

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

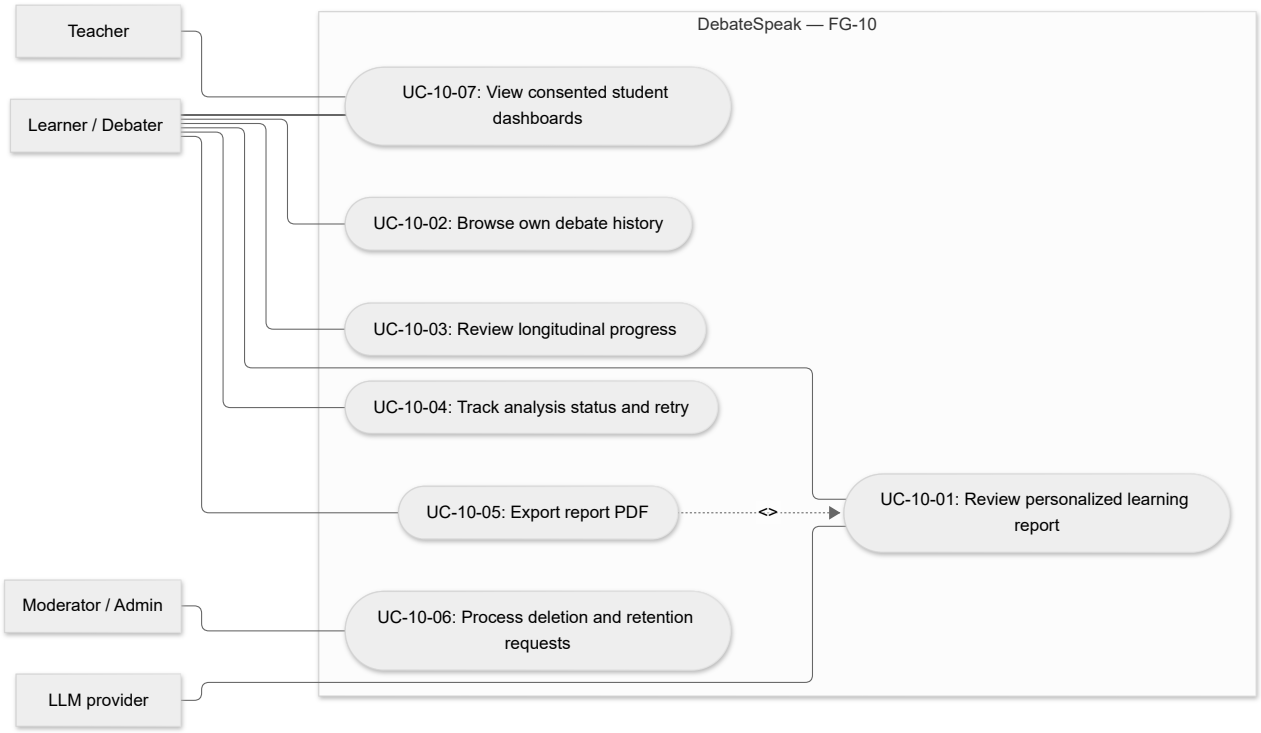


ID	User goal / supporting behavior	FRs	Delivery scope
UC-09-01	Review debate evaluation and consensus	FR-44, FR-45	future-mvp
UC-09-02	Configure analysis and inspect usage	FR-46	should
UC-09-03	Run optional heterogeneous judge panel	FR-47	stretch
UC-09-04	Assign a human evaluator	FR-55, FR-56	should
UC-09-05	Submit human grading card	FR-55, FR-56	should
UC-09-06	View shared human grades	FR-55, FR-56	should

The baseline debate score is a single-judge evaluation with self-reported confidence capped at 0.8. Multiple claim evaluators do not imply a heterogeneous debate-judge panel. Optional human grades are separate records available only for an assigned ENDED debate under dual sharing consent; the UI identifies the evaluator and rubric/card versions.

FG-10 — History and reports

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).



ID	User goal / supporting behavior	FRs	Delivery scope
UC-10-01	Review personalized learning report	FR-48, FR-49	future-mvp
UC-10-02	Browse own debate history	FR-50	future-mvp
UC-10-03	Review longitudinal progress	FR-51	should
UC-10-04	Track analysis status and retry	FR-52	should
UC-10-05	Export report PDF	FR-53	stretch
UC-10-06	Process deletion and retention requests	FR-54, FR-04	future-mvp
UC-10-07	View consented student dashboards	FR-57	post-mvp

Reports expose unavailable analysis rather than inventing sections or praising unsupported performance. Trends are observations, not certified proficiency gains. Teacher dashboards are read-only and revocable; class creation, invitations, rosters, assignments, group chat and multi-debater room design remain excluded long-term goals.

6. Requirement-to-use-case traceability

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

Every FR in the architecture's current FR-01–60 set appears below. A mapping means the behavior is modelled and specified; it is not evidence that the feature is implemented. Current build scheduling comes from the roadmap and PA3 scope.

Requirement	Use cases
FR-01	UC-01-01, UC-01-02
FR-02	UC-01-03

Requirement	Use cases
FR-03	UC-01-04, UC-01-05, UC-01-06
FR-04	UC-01-07, UC-10-06
FR-05	UC-01-08
FR-06	UC-02-01
FR-07	UC-02-02
FR-08	UC-02-03
FR-09	UC-02-04
FR-10	UC-03-01
FR-11	UC-03-02
FR-12	UC-03-01, UC-03-02, UC-03-03, UC-03-05, UC-03-06, UC-03-13
FR-13	UC-03-04, UC-03-05, UC-03-13
FR-14	UC-03-07
FR-15	UC-03-06, UC-04-03
FR-16	UC-03-08
FR-17	UC-03-09
FR-18	UC-04-02
FR-19	UC-03-05, UC-03-13, UC-04-01
FR-20	UC-04-02
FR-21	UC-04-03
FR-22	UC-04-04
FR-23	UC-05-01
FR-24	UC-05-01
FR-25	UC-05-01
FR-26	UC-05-01
FR-27	UC-05-02
FR-28	UC-05-03
FR-29	UC-06-02
FR-30	UC-06-01, UC-06-05
FR-31	UC-06-01
FR-32	UC-06-03
FR-33	UC-06-04
FR-34	UC-07-01
FR-35	UC-07-01
FR-36	UC-07-01
FR-37	UC-06-05, UC-07-02
FR-38	UC-08-01
FR-39	UC-08-01
FR-40	UC-08-01
FR-41	UC-08-01
FR-42	UC-08-02
FR-43	UC-08-01
FR-44	UC-09-01
FR-45	UC-09-01
FR-46	UC-09-02
FR-47	UC-09-03
FR-48	UC-10-01
FR-49	UC-10-01
FR-50	UC-10-02
FR-51	UC-10-03
FR-52	UC-10-04
FR-53	UC-10-05
FR-54	UC-10-06
FR-55	UC-09-04, UC-09-05, UC-09-06
FR-56	UC-09-04, UC-09-05, UC-09-06
FR-57	UC-10-07
FR-58	UC-03-10
FR-59	UC-03-11
FR-60	UC-03-12

7. Cross-cutting acceptance and human consequences

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107).

- Consent is explicit, uncoerced and separately controlled by each debater; signing in, joining, clicking READY or granting browser microphone permission does not grant transcription/analysis consent.
- Refusing consent preserves normal account use and an accessible leave path. Before dual consent, a microphone test remains local, without upload, storage or transcription.
- Live controls prioritize audibility, mute, presence and leaving. Detailed critique belongs in post-debate reflection; optional AI failures cannot stop healthy audio.
- Capture stops promptly on end or consent withdrawal. Reconnection never creates a duplicate seat or silently continues capture after authorization is lost.
- Participants can access only authorized room/transcript/report data. Human evaluators see assigned ended debates only; teachers have explicit revocable read-only access.
- Every retained language/argument feedback quote belongs to the correct speaker and segment. Feedback states a concrete issue and remediation without insulting the learner or assigning motives.
- Raw audio is not stored by default. Erasure/retention workflows cover derived analyses and optional chat as well as the primary transcript.

8. Review and completion evidence

Performed: Trần Lê Anh Tuấn (24125107), AI Orchestrator & Analysis Lead. Reviewed: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead, 2026-11-14. Edited: Trần Lê Anh Tuấn (24125107); Nguyễn Bảo Minh Triết (24125047), 2026-11-14.

The inventory contains **54 use cases across ten functional groups**, including two mandatory supporting subcases. All 60 FRs are mapped. There is one overview and ten detailed Mermaid diagrams. Part D describes every catalogue ID and links its prototype and alternative-flow evidence. use_cases.json enables automated cross-checking of IDs, actor references, requirements and screenshot coverage.

Human reviewer: Nguyễn Hồng Tấn Tài (24125078), Backend & Real-Time Media Lead. Review completed 2026-11-14. Diagram rendering, PDF compilation and document-validation results are recorded in the overall local work record after execution; no successful check is inferred solely from this model's existence. The model is merged into `pa3`. On 2026-11-14 delivery scopes were revised to the PA3 implementation: FG-01 is Part E, and UC-03-01 to UC-03-03 move to PA4.